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Bacteriophages for ESKAPE: Role in pathogenicity and measures of control

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Abstract

Introduction: The quest to combat bacterial infections has dreaded humankind for centuries. Infections involving ESKAPE (*Enterococcus* spp., *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter* spp.) impose therapeutic challenges due to the emergence of antimicrobial drug resistance. Recently, investigations with bacteriophages have led to the development of novel strategies against ESKAPE infections. Also, bacteriophages have been demonstrated to be instrumental in the dissemination of virulence markers in ESKAPE pathogens.

Areas covered: The review highlights the potential of bacteriophage in and against the pathogenicity of antibiotic-resistant ESKAPE pathogens. The review also emphasizes the challenges of employing bacteriophage in treating ESKAPE pathogens and the knowledge gap in the bacteriophage mediated antibiotic resistance and pathogenicity in ESKAPE infections.

Expert opinion: Bacteriophage infection can kill the host bacteria but in survivors can transfer genes which contribute towards survival of the pathogens in the host and resistance towards multiple antimicrobials. The knowledge on the dual role of bacteriophages in the treatment and pathogenicity will assist in the prediction and development of novel therapeutics targeting antimicrobial resistant ESKAPE. Therefore, extensive investigations on the efficacy of synthetic bacteriophage, bacteriophage cocktails and bacteriophage in combination with antibiotics are needed to develop effective therapeutics against ESKAPE infections.

Keywords: Bacteriophage, lysogenic cycle, pathogenicity, lytic cycle, treatment, ESKAPE

Article highlights

- Bacteriophages are considered as an important strategy to treat ESKAPE infections that confer antimicrobial resistance.
- Under certain environmental cues, bacteriophage can enhance the virulence property of ESKAPE pathogens.
- Understanding the molecular mechanism of the interactions between bacteriophage, bacteria and humans can lead to the development of novel therapeutic strategies against ESKAPE infections.

1. Introduction

Antimicrobial resistance (AMR) is a multisectoral global health concern that needs immediate action for its containment [1]. Research efforts directed towards a comprehensive understanding of AMR and virulence mechanisms are the reliable key for developing novel therapeutic strategies to avoid the emergence and spread of AMR. Recent investigations have reported the role of bacterial quorum sensing [2], biofilms, CRISPR-Cas system[3], and host metabolites [4] in the emergence of virulence assisted by AMR. Currently, several antibiotics [5] or combinational therapies such as antimicrobial compounds [6], peptides, extracellular proteases[7], lactones [8], quorum sensing inhibitors [9–11], antibacterial or antibiofilm antibodies, phytochemicals [9,10], antibiotics conjugated with the strategies mentioned above are being explored and instigated. However, in recent years, the use of bacteriophages to overcome antimicrobial resistance in ESKAPE infections and the implications associated with it are under investigation [12]

Bacteriophages are viruses, which specifically infect bacterial cells and are one of the most abundant organisms in the environment (~total population 10^{30}–10^{32}). Bacteriophage possesses tremendous diversity in morphology and genomic content enclosed in its capsid. The basal plate of the bacteriophage consists of tail fibers required for the initial attachment to the host cells. The bacteriophage genetic material is introduced to the host cell via the sheath region, often surrounded by sheath proteins [13]. Inside the host bacterial cell, bacteriophage undergoes either lytic (virulent) or lysogenic (non-virulent) life cycle. The bacteriophage is dependent on the host machinery for the synthesis of bacteriophage particles, assisted by early proteins synthesized during the lytic cycle. The genetic material is then packaged into the capsid constructing a new daughter bacteriophage that eventually bursts into the surrounding after lysis of the host bacterial cell. However, during the lysogenic cycle, rather than undergoing lysis, bacteriophage persists in the host in a dormant state. Bacteriophages and hosts in the lysogenic cycle are known as prophage and lysogen, respectively [14]. Based on the environmental cues, such as changes in pH, nutrients, temperature and exposure to foreign DNA, hydrogen peroxide, antibiotics or other agents causing DNA damage prophage enter into a lytic cycle and lead to lysis of the host cell [15] (Figure 1)

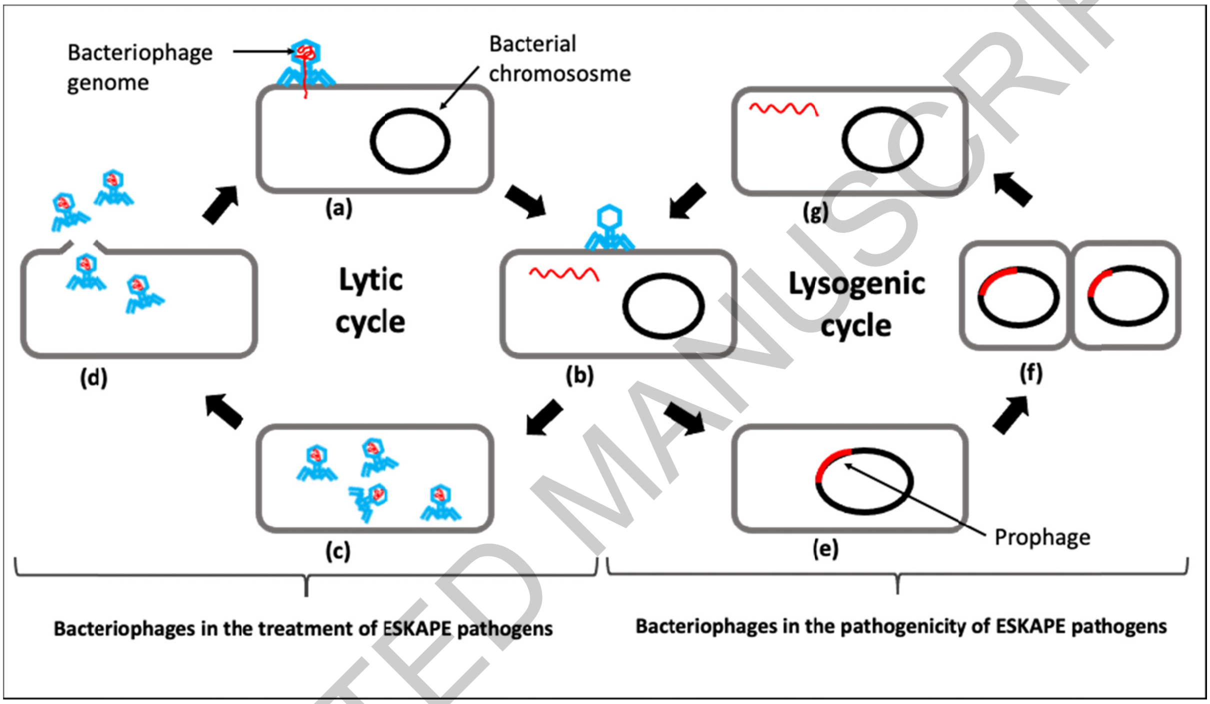

Figure 1. The life cycles of bacteriophage: (a) Bacteriophage binds to bacterial cell (b) Bacteriophage injects genome in the bacterial cytoplasm (c) The synthesis of new bacteriophages (d) Bacterial cell lysis and release of new phage (e) Bacteriophage genome integrates into the bacterial chromosome. Bacteriophages and hosts in the lysogenic cycle are called as prophage and lysogen, respectively. (f) Prophage DNA is copied during cell division (g) The excision of the bacteriophage genome from the bacterial chromosome

Recently, the majority of the research has been focused on bacteriophage and host interactions. Several studies have revealed that the lytic and lysogenic cycles confer unique phenotypes to the bacteriophage and their bacterial host. The usage of lytic bacteriophages in treating bacterial infections is gaining importance due to its efficacy towards antibiotic-resistant bacterial infections [12]. Among such antibiotic-resistant infections, the highest-burden is imposed by the 'ESKAPE' pathogens. The ESKAPE infections are difficult to treat since they depict phenotypes such as multidrug resistance (MDR), extended drug resistance (XDR), and Pandrug resistance (PDR) [16–18]. The potential role of bacteriophage therapy against these drug-resistant infections is currently under investigation. Also, the lysogenic cycle of the bacteriophage is being explored as it contributes to bacterial pathogenicity. Hence, comprehensive knowledge of the role of bacteriophages in bacterial pathogenicity is vital for the development of ideal therapeutic strategies against ESKAPE infections.

Moreover, bacteriophage demonstrates a dual role in bacterial infections. Bacteriophage can enhance the virulence properties of the bacteria as well as can inhibit their growth. This review summarized the recent articles in investigating the potential role of bacteriophage in the pathogenicity of ESKAPE pathogens (all the previous reports) and against the pathogenicity of ESKAPE pathogens (from the year 2015 to 2020). A literature search was done using the PubMed database. For articles on bacteriophage therapy in ESKAPE infection control, search terms included 'organism (Enterobacter spp./ *Staphylococcus aureus*/ *Klebsiella pneumoniae*/ *Acinetobacter baumannii*/ *Pseudomonas aeruginosa*/ *Enterobacter spp.*) and bacteriophage therapy'. Moreover, for the articles on bacteriophages inducing ABR in ESKAPE, search terms include 'Bacteriophages and pathogenicity in the organism (Enterobacter spp./ *Staphylococcus aureus*/ *Klebsiella pneumoniae*/ *Acinetobacter baumannii*/ *Pseudomonas aeruginosa*/ *Enterobacter spp.*). The review also emphasizes the challenges of exploiting bacteriophage in the treatment of ESKAPE pathogens.

2. Bacteriophages in the treatment of ESKAPE infections

Antibiotics are widely used for treating bacterial infections since their discovery in 1928 [19]. Currently, antibiotics have become an integral part of modern medicine such as sepsis treatments, surgeries, treatments of infections associated with the chronic condition (e.g. diabetes), organ transplant, dialysis for severe kidney diseases and cancer treatments. Hence, antibiotic resistance is one of the biggest threats to global health [20] and alternative or combination treatment strategies are required to develop immediately [12]. In recent years, bacteriophage therapy is being explored as a potential treatment strategy [12]. Unlike antibiotics, bacteriophages are imparted with characteristics of self-amplification, host specificity, less toxicity and disintegrate established biofilms. [21]. Bacteriophage therapy includes treating bacterial infections by mono-bacteriophage, bacteriophage cocktail, bacteriophage protein, and bacteriophage combined with antibiotics. Several *in vitro* and *in vivo* studies on the efficacy of bacteriophage therapy on ESKAPE pathogens have been described in this review (Table 1).

2.1. *Enterococcus* spp

Enterococci are pathogenic organisms causing urinary tract infections, endocarditis, bacteremia and many more. Enterococci possess intrinsic resistance to antibiotics e.g. β-lactam- antibiotics, aminoglycosides. In addition, enterococcal biofilms impede penetration of the antibiotics through the biofilm matrix and contribute to antibiotic resistance [22]. Hence, treating Enterococci infection is a challenge due to biofilm formation and a rise in drug resistance. However, studies report that bacteriophage EFDG1 [23], vB_EfaS-Zi, vB_EfaP-Max [24], EF1TV [25] and a cocktail of EFDG1 and EFLK1 in the ratio 1:1 [26] can effectively decrease the *Enterococci* load within the biofilm. Moreover, an engineered bacteriophage φEf11/φFL1C (Δ36) [27], bacteriophage EFp29 [28], bacteriophage HEF13

[29], bacteriophage vB_EfaS_LM99 [30], bacteriophage cocktail (EFDG1 and EFLK1) with an antibiotic (12.5 mg/kg of ampicillin) [31] and the chimeric lysine-ClyR [32] demonstrated growth inhibition of *Enterococcus* by ≥ 90% (Table 1). Similarly, an *in vivo* root canal infection model confirmed the efficacy of the bacteriophage EFDG1 and EFLK1 formulated with Kolliphor P 407 (poloxamer) in the treatment of *Enterococcus faecalis* infection [33]. Endophthalmitis is one of the severe infections caused by *Enterococcus spp.* It needs immediate treatment as the progression of the infection is rapid and often lead to permanent vision loss. The available treatment strategies have limitations as they lead to the infiltration of inflammatory cells and also cause retinal detachment. However, bacteriophage ΦEF24C-P2 treatment in the endophthalmitis mouse model showed rapid lysis of bacterial cells without inducing inflammatory response or retinal toxicity in the mouse model (Table 1) [34].

2.2. Staphylococcus aureus

S. aureus is an opportunistic pathogen. However, it is also a major etiological agent in bloodstream infections like sepsis [35]. A study reported 90-100% inhibition in the growth of clinical isolates of *S. aureus,* both in the broth and on the abiotic surfaces, by using a bacteriophage cocktail of M1, M1M4, CJ17, CJ12 [36]. The first phase-I clinical trial of bacteriophage cocktail AB-SAO1(Sa83+ Sa87+J-Sa36) was carried out in patients with chronic rhinosinusitis. The study showed intranasal administration of AB-SAO1 is safe and can be used as an alternative to antibiotics [37]. The intranasal administration of bacteriophage cocktail AB-SAO1 in a mouse infection model of acute pneumonia displayed a 94-95% decrease in the count of clinical *S. aureus* isolate (Table 1). This preclinical data support a phase-I clinical trial of AB-SAO1 bacteriophage cocktail [38] (Table 1). Moreover, intravenous administration of bacteriophage cocktail AB-SAO1 was effective in patients with severe endocarditis and septic shock. The safety and tolerability of the AB-SAO1 were assessed and no bacteriophage resistance was reported during the study [39,40].

Likewise, bacteriophage-based drug 'Staphal' exhibited inhibition in the growth of the *S. aureus* strain in the planktonic and biofilm when studied *in-vitro* using catheter disc method and microtiter plate method. Nevertheless, the destruction of the bacterial cells within biofilm required a higher concentration of the bacteriophage as compared to the planktonic stage [41]. Another study reported the efficacy of bacteriophage LM12 and Sta bacteriophage cocktail against *S. aureus* strain isolated from orthopedic implant-associated infection. The bacteriophage and the cocktail inhibited the growth of *S. aureus* by 90-100% when treated *in vitro* [30,42]. An *in vivo* study compared the efficacy of mono- bacteriophage therapy and antibiotic (8mg/kg clindamycin) therapy distinctly and in combination. The data revealed that the intravenous injection of mono-bacteriophage was relatively more effective (100% bacterial reduction) than antibiotic (62-87% bacterial reduction) or the combination treatment (75-90% bacterial reduction) [43]. In contrast, combination therapy of bacteriophage Sb-1 and antibiotic (fosfomycin/ rifampin/ vancomycin/ daptomycin/ ciprofloxacin) effectively eradicated *S. aureus* ATCC 43300 biofilm formation and reduced bacterial burden *in vitro* by 75-90% [44] (Table 1).

Moreover, intranasal/ subcutaneous administration of bacteriophage proteins like LysGH15 (from bacteriophage GH15) and apigenin [45], topical application of LysGH15-api-Aquaphor [46], intravenous administration of Lysin P128 [47], or intranasal administration of Lysin Sal200 [48] displayed ≥ 90 % inhibition in the growth of *S. aureus* USA300 under *in vivo* conditions. The bacteriophage proteins LysGH15-api-Aquaphor and Lysin Sal200 showed no significant increase in the inflammatory cytokines such as IL-1, TNF-α, and IFN-γ [46,48] (Table 1).

Furthermore, an *in vitro* study stated the role of Ca^{2+} and Zn^{2+} in increasing the lytic activity of the lysin protein Lys-phiSA012 against *S. aureus* by 80-90% [49]. However, the combination of the bacteriophage and bacteriophage protein endolysin vB_SauM-LM12 (LM12) exhibited a 90-100% reduction in the growth *S. aureus* in the suspended state during the stationary phase. In contrast, the combination was relatively ineffective against the biofilm [50] (Table 1).

2.3. Klebsiella pneumoniae

The gram-negative pathogen, *K. pneumoniae*, causes nosocomial infections, including pneumonia, wound infections, meningitis. A study revealed a reduction in the growth of *K. pneumoniae* MTCC 109 by 80-100% on the intranasal administration of the bacteriophage in the pneumonic BALB/c mouse model [51]. Similarly, the cocktail treatment involving bacteriophage ECP311, KPP235, and ELP140 when injected into the last left proleg of *Galleria mellonella* inhibited *K. pneumoniae* (kp235), displaying a 100 % survival rate of the larva. However, the *in vivo* model required multiple bacteriophage doses to achieve a successful treatment process [52,53]. Bacteriophage TSK1 demonstrated a 99-100% reduction in the *in-vitro* developed *K. pneumoniae* biofilm by producing a depolymerase enzyme and targeting the capsular polysaccharides present in the biofilm [54]. Moreover, the efficacy of polyvalent bacteriophage preparation was studied using the mouse bacteremia model. Bacteriophage cocktail active against 6 pathogens including *K. pneumoniae* (10 isolates), *E. coli* (10 isolates), *H. influenzae* (4 isolates), *P. aeruginosa* (3 isolates), *M.*

catarrhalis (1 isolate) and *C. freundii* (1 isolate) showed efficacy after 45 min of injection of bacterial pathogens [55] (Table 1).

2.4. Acinetobacter baumannii

The pathogenic bacteria *A. baumannii* can cause infections of the urinary tract, wound infections and bacteremia. An *in vivo* study demonstrated complete eradication of the *A. baumannii* strain at the wound site by topical application of the bacteriophage vB-GEC_Ab-M-G7 in rat wound infection model [56]. However, bacteriophage Bφ-C62 showed MOI (multiplicity of infection) dependent efficacy against *A. baumannii* under *in vitro* and *in vivo* conditions [57]. Moreover, intranasal treatment with bacteriophage vB_AbaM-IME-AB2 [58] and intraperitoneal administration of bacteriophage φkm18p [59] completely rescued the mice with *A. baumannii* infection and depicted a decrease in inflammation in the lung of mice, followed by reduced levels of TNF-α and IL-6 [58]. However, despite the low MOI, subcutaneous or direct injection of the bacteriophage Abp1 in the wound site of the mouse model displayed a 100% healing of the wound-induced by *A. baumannii* [60]. Likewise, complete eradication of the *A. baumannii* was observed by bacteriophage Bφ-R2096 administrated through the nasal route in mice. In contrast, a 50% reduction in the growth of bacteria was observed in *G. mellonella,* administered via proleg injection [61]. On the other hand, a bacteriophage cocktail consisting of vB_AbaM_30 54 and vB_AbaM_30 90 eradicated *A. baumannii* infection in both the mouse and *G. mellonella* [62]. Similarly, a bacteriophage cocktail containing vB_AbaS_D0 and B_AbaP_D2 demonstrated the ability to reduce bacteremia induced in both *in vivo* and *in vitro A. baumannii* infection models. The efficiency of the bacteriophage cocktail was significant than mono-bacteriophage therapy [63] (Table 1).

2.5. Pseudomonas aeruginosa

P. aeruginosa is a gram-negative bacteria causing a broad range of diseases such as respiratory tract infections, urinary tract infections and skin infections. It was observed that the bacteriophage cocktail DL52, DL60, DL62, DL64, and DL68 employed in the eradication of *P. aeruginosa* biofilm in the static and dynamic model exhibited 95% and 100% eradication of the biofilm respectively [64]. However, the bacteriophage cocktail injected into the hemolymph of *G. mellonella* within 15 hours of infection exhibited an 85-100% reduction in the growth of *P. aeruginosa* [65]. As compared to the individual bacteriophages, the cocktail consisting of PYO2, DEV, E215, E217, PAK_P1 and PAK_P4 showed a broader host range. The single bacteriophage with the highest host range lysed 62% of *P. aeruginosa* strains in the collection and 55% of the clinical strains isolated from cystic fibrosis patients. However, the phage cocktail lysed nearly 77% of *P. aeruginosa* strains in the collection and 75% of the clinical strains that are isolated from cystic fibrosis patients. In addition, the same cocktail (MOI of 0.05 for each bacteriophage) when used in the treatment of *P. aeruginosa* respiratory infection in mice through nasal route, it showed 100% survival in mice [67]. However, *P. aeruginosa* respiratory infection in mice treated with nasal administration of PAK-P1 bacteriophage required a 1:1 ratio of phage to bacterial cells to achieve 100% survival in mice [66]. Conversely, the cocktail failed to attain complete eradication of the pathogen in the *G. mellonella* infection model. However, prophylaxis with the bacteriophage cocktail 1 hour before the bacterial challenge protected *G. mellonella* from *P. aeruginosa* infection [67] (Table 1).

2.6. Enterobacter spps.

Among the *Enterobacter* group, *Enterobacter cloacae* are considered as an important nosocomial pathogen responsible for various infections, including gastrointestinal infections, skin and soft tissue infections, urinary tract infections, bacteremia, and endocarditis. It has been reported that injection of 3 doses of either individual bacteriophage ELP140 or bacteriophage cocktail comprising of ECP311, KPP235 and ELP140 into the hemolymph of *G. mellonella* at an interval of 6 hours could eradicate 100% of the bacteria [52]. Moreover, individual bacteriophage and bacteriophage cocktail inhibited the growth of *E. cloacae* associated with the urinary tract infection by ≈3 log and ≈4 log respectively [68] (Table 1).

In conclusion, individual bacteriophages, bacteriophage cocktails, bacteriophage proteins, engineered bacteriophages alone and in combination with antibiotics have excellent potential in the treatment of ESKAPE pathogens. However, the efficiency of each strategy varies depending on the inherent and acquired properties of the bacteriophage and bacterial strain. Hence, further studies are still required to avoid limitations of the bacteriophage therapy.

3. Limitations of bacteriophage therapy

Although several studies have demonstrated the role of bacteriophage in eradicating infections, bacteriophage therapy against ESKAPE has several concerns. The precise pre-diagnosis of bacterial infections, such as identifying each bacterial species at the infection site, is essential for effective and personalized bacteriophage therapy. However, the diagnosis of ESKAPE infections is challenging [69]. The activation of immune responses like the production of anti- bacteriophage antibodies and mononuclear phagocytes may neutralize the bacteriophage [70]. Furthermore, the intracellular pathogens are difficult to eradicate as pathogens are not accessible by a bacteriophage, thereby evading the therapy [71]. Also, the lytic capacity of the bacteriophage may differ due to the interaction between the bacteriophage and pathogen [72]. Besides, bacteriophages without an integrase gene, antibiotic resistance genes and bacteriophage toxin genes are difficult to produce [73,74]. ESKAPE pathogens are profoundly associated with polymicrobial infections and commonly require bacteriophage cocktail or combination therapy than mono- bacteriophage therapy (Table 1). However, the safety and stabilization of bacteriophages in the cocktail and combination therapy is an intricate process [72]. Another major concern with bacteriophage therapy is the development of resistance towards bacteriophages in bacteria [75]. Though, a study reported that the efficacy of bacteriophage therapy could be increased by coevolution, which is achieved by interrupting the resistance evolution in *P. aeruginosa* using other bacteriophage. In this process, the bacteriophages are prior adapted to bacterial strains to evolve and increase efficacy compared to non-adapted bacteriophages [76].

There are numerous reports on the bacteriophages against ESKAPE. However, due to the aforementioned limitations, this strategy is not yet accepted and commercialized.

4. Bacteriophages in the pathogenicity of ESKAPE

The phenotype of the bacteria is often governed by the extrachromosomal genetic components such as genomes of plasmids and bacteriophages. The manipulation through such genetic elements leads to increased resistance to antibiotics and enhanced pathogenicity through toxin synthesis. Several virulence factors encoded within the bacteriophage genome can regulate the virulence properties of the bacteria. In recent years, prophages have been the focus of research, as they deliver myriad advantages to the bacterial host (lysogen), such as increased resistance to antibiotics, virulence and overall fitness of the bacterial host [77]. Prophages possess cargo genes known as morons involved in the virulence potential of host bacteria [78]. Relative genomics studies have revealed that the genomes of bacteria and their bacteriophages are coevolving, leading to increasing the pathogenicity of bacteria by bacteriophage [78].

4.1. *Enterococcus* spp.

The *Enterococcal* strains associated with root canals and subgingival plaque frequently possess prophage ϕEf11 (ϕEf11 bacteriophage genome is integrated into the genome of *Enterococcus spp.*), and it has been observed that the prevalence of prophage can enhance biofilm formation, increased persistence at high cell density and also the resistance towards lytic enzymes of bacteriophage [79]. The presence of multiple prophages, polylysogeny, plays a prime role in bacterial evolution, contributing to the fitness and production of virulence factors. *E. faecalis* V583 is one of the high-risk clinical isolates comprising 7 prophages (V583-pp1 to V583-pp7). A study revealed that *E. faecalis* V583 prophages encode platelet binding like proteins that facilitate adhesion of the pathogen to human platelets, thereby increasing the risk of endocarditis infection. Prophages play an important role in the *E. faecalis* pathogenicity [80]. Moreover, sequence analysis of the bacteriophages isolated from the clinical isolates of *E. faecalis* that were purified by pulsed-field gel electrophoresis, followed by restriction mapping, revealed genes contributing to the virulence and fitness of the strain [81].

4.2. Staphylococcus aureus

The *S. aureus* bacteriophage genome confers genes responsible for several virulence factors such as Panton-valentine leukocidin (*lukSF)*, chemotaxis inhibiting proteins (*chp*), staphylokinases (*sak*), exfoliative toxins (*eta*), and enterotoxins (*sea*). They also assist the transfer of pathogenicity island (SaPIs) to the new bacterial host. Thus, bacteriophages impart diversity among the *S. aureus* strains in terms of pathogenicity.

A strong association has been reported between the expression of lysogeny module, lytic module and virulence factor, which influence the expression of pathogenicity genes during bacteriophage induction. The lysogeny module encodes integrase (Int, CI) and regulator proteins (Cro). The transition between the lysogenic and lytic life cycle is determined by the expression of int, CI and Cro proteins. The Int protein assists in the integration of the phage genome in the bacterial chromosome. The expression of CI protein maintains a lysogenic state. However, the expression of Cro triggers the lytic cycle of the phage. Prophage with int gene has been recently found in the epidemic methicillin-resistant *Staphylococcus aureus* strain. Also, the two-component regulatory system (quorum-sensing system) is found to be involved in the expression of phage-encoded virulence factors such as eta, pvl, scn, and chp. It indicates a close association between the phage life cycle and bacterial virulence. It has been also observed that phage inducing conditions such as exposure to UV, reactive oxygen species, antibiotics, and change in nutrients, pH, temperature increases transcription of the virulence factors that are present in proximity to the lysis module of the phage genome. The methicillin-resistant *S. aureus* isolated from the skin or soft tissue infection, and necrotizing pneumonia, indicated the role of the *lukSf* comprising bacteriophages in the pathogenicity of *S. aureus*. Generally, most of the bacteriophages consist of one virulence factor. However, certain bacteriophages like phiSa3 and phiN315 codes for almost 5 virulence factors. *S. aureus* bacteriophages Sa3int, codes for immune evasion cluster and immune-modulatory proteins such as Chips, Sak, Sea, and Scin that may support the colonization of the strain regardless of the innate immune response [82,83]. Moreover, *S. aureus* bacteriophages mediate the transfer of SaPIs encoding for toxins and contribute to enhanced pathogenicity of the recipient bacterial cell. The bacteriophage dynamics in the form of transfer of the bacteriophage, duplication, and stable extrachromosomal integration generate heterogeneity within the infecting bacterial population, by regulating the virulence factors and thus increase the pathogenic potential of the whole bacterial consortium [83].

Furthermore, a sequencing study illustrated the presence of prophage in clinical isolates of *S. aureus* CC398. Ancestor naïve cells of *S. aureus* CC398 which were devoid of prophage can colonize only the animal cells. However, prophage containing *S. aureus* CC398 can affect both humans and animals. This indicates the potential role of prophage in the transfer of human pathogenesis phenotype (such as the ability to bind to human extracellular matrix proteins and to enter and live within non-phagocytic cells) to naïve *S. aureus* CC398 cells [84]. Also, an RNA seq data of *S. aureus* RN450 and *S. aureus* RN450 biofilms, encompassing prophages ϕ80α and ϕ11, exhibited a difference in their gene expression profiles. Biofilms of prophage carrying strain displayed elevated expression of the SigB regulon and decreased expression of the genes regulated by the Agr quorum sensing systems and protease encoding genes, responsible for bacterial dispersion. Though prophage carrying strains presented increased biofilm formation, further studies are required to understand the role of the two prophages in establishing an infection [85]. Certain bacteriophage play a key role in the inter-genus (e.g., *S. aureus* to *Listeria monocytogenes* in raw milk) as well as the intra-genus transfer of SaPI [86]. SaPI encodes its repressor Stl that possesses several domains to interact with the bacteriophage proteins. Bacteriophage proteins bind to Stl repressor and initiate SaPI transfer. The mechanism for the transfer of SaPI demonstrates the role of bacteriophage in inducing pathogenicity of the recipient strain of other genera [87]. Similarly, another study revealed the role of the prophages in the induction of SaPIs in 7 clinical isolates of *S. aureus* and methicillin-resistant *S. aureus* strain. MRSA carrying ϕSa2mw and ϕSa3mw bacteriophages assist the transfer of SaPImw2 pathogenicity island encoding *sec*, *sel* and *ear* enterotoxins by binding to Stl-DUF3113 [88].

4.3. Klebsiella pneumoniae

K. pneumoniae (CVUAS 5665-2) isolated from the mastitis milk sample comprises of bacteriophage KPP5665-2, which possesses characteristics of the *Siphoviridae* family. However, further studies are required to correlate its role in the pathogenesis of the bacteria, causing mastitis [89]. A study screened the clinical isolates of *K. pneumoniae* 675920, 721005, 911021 and 19051 for the prevalence of prophages at *sap* sites to decipher their relevance with the horizontal transfer of resistance genes, to elucidate molecular mechanisms of antimicrobial resistance in the pathogen [90]. It has been reported that *K. pneumoniae* possesses both virulence and resistance genes. A genome analysis study described that the drug-resistant *K. pneumoniae* are more diverse and more likely to receive virulence genes from mobile genetic elements like prophages as compared to virulent *K. pneumoniae* [91]. Similarly, a study depicted the prevalence of P2-P4 prophages in the *K. pneumoniae* (CG258). However, the CRISPR- Cas system had a limited effect on the prevalence of P2-P4 prophages in the CG258. Furthermore, the authors also demonstrated the contribution of prophages in the chromosomal integration of AMR genes and thereby to the fitness of CG258 [92].

4.4. Acinetobacter baumannii

Along with the development of new antibiotics, understanding the underlying mechanism of AMR and the evolution of pathogenicity is equally important. The genome of the *A. baumannii* consists of a small core genome and a large diverse accessory genome. It was observed that the genes corresponding to AMR are situated on the core as well as the accessory genome. However, in the accessory genome it is flanked by integrases transposes. This data suggests the probability of horizontal gene transfer through plasmid or bacteriophages. The prophages are found to play a crucial role in the evolution of pathogenicity of *A. baumannii*. It was also reported that *A. baumannii* prophages contain genes encoding for the production of virulence as well as other pathogenic phenotypes such as host colonization, AMR, host immune evasion and survival in extreme environmental conditions [77]. The sequencing of the MDR strain of *A. baumannii* (R2090) exhibited the prevalence of pathogenicity and AMR genes (*bla*NDM-1 carbapenemase gene) providing insights into the horizontal gene transfer mechanism [93]. This indicates the vital role of the mobile genetic elements like bacteriophage or plasmid in the pathogenicity of the R2090 strain. Also, the mechanism was validated through artificially enclosing eDNA in the bacteriophage capsid, transducing bacteriophage-eDNA assembly to naïve *A. baumannii* without intercellular interaction and further assessing the presence of AMR genes in the naïve recipient cells [94].

4.5. Pseudomonas aeruginosa

It has been demonstrated that *P. aeruginosa* Liverpool Epidemic Strain LESB58 comprises of 6 prophages. The transcription profile revealed the signature tagged mutations in the prophages, affecting the establishment of infection by altering the virulence gene expression, indicating a role of the bacteriophages in the pathogenicity of *P. aeruginosa* [95]. A detailed evolutionary study predicted that the bacteriophages altered the mode of bacterial evolution when *P. aeruginosa* population was allowed to replicate and evolve in the presence and absence of 3 temperate bacteriophages representing the cystic fibrosis lung infection model. Though the integration of the phage ϕ4 in the bacterial genome is random, specific genes like motility associated genes and quorum sensing regulatory systems are particularly selected, suggesting that insertional inactivation of these genes are a part of adaptive evolution [96]. Similarly, Pf4 prophage plays a vital role during the several steps of *P. aeruginosa* biofilm formation and increases bacterial virulence [97] Also, filamentous prophages are predominantly observed in the environmental and clinical isolates of *P. aeruginosa*. Pf4 and Pf5 prophages have been reported to integrate into the *P. aeruginosa* PAO1 and *P. aeruginosa* Pf5, respectively, contributing to biofilm development. The XisF4 and XisF5 excisionases are also involved in switching the lysogenic cycle to the lytic cycle of the bacteriophage. Moreover, it has been observed that XisF4 triggers the replication of the Pf4 bacteriophage. Additionally, *xisF4* and its adjacent bacteriophage repressor gene *pf4r* are transcribed separately. Pf4r and XisF4 activate and repress the expression of each other, where MvaT and MvaU (H-NS family proteins) repress xisF4, further suppressing the production of Pf4 filamentous bacteriophage [98]. Another study demonstrated the role of the bacteriophage F116 of *P. aeruginosa* in the genetic exchange among two species on the plant [99]. The role of the expression of the bacteriophage morons on the virulence phenotype of the *P. aeruginosa*, PAO1 and PA14 was studied using a library of the bacteriophages. Bacteriophage moron was found to be involved in the swimming and twitching motility and the expression of virulence factors, indicating its contribution to the human disease. However, bacteriophage mediated phenotypic changes still need further investigation [100]. It was also observed that *P. aeruginosa* bacteriophage Ab31 assists bacteria during the colonization of the lung in cystic fibrosis patients by expressing genes required for the adaptation and persistence of the pathogen [101]. The sequence analysis of *P. aeruginosa* (BH9) using the ion proton system and long-read PacBio SMRT platform displayed the prevalence of the *bla*KPC-2 gene and Mu like bacteriophage in the pBH6 plasmid. Bacteriophage plasmid complex co-integrated into *P. aeruginosa* BH9 increases AMR and pathogenicity [102]. Furthermore, *P. aeruginosa* bacteriophage diversity analysis revealed that Pf1, like prophages, is highly prevalent (60%) in the *P. aeruginosa*. However, the role of such bacteriophages in the virulence of pathogens is necessary to be explored [103]. Also, Mu like transposable bacteriophage such as LESϕ4 can induce divergence in the siderophore production by *P. aeruginosa* strain grown under iron-limiting and iron-rich environment, suggesting a higher frequency of the co-existence of siderophore producers and siderophore nonproducer in iron limiting conditions [104].

4.6. Enterobacter spp

A genomic analysis of the *Enterobacter hormaechei* outbreak strain (EHOS) in The Netherlands, exhibited an increase in the number of genes related to a transferable element, including prophages in the genome. This suggested the genome plasticity of *E. hormaechei* [105]. Moreover, genomic data analysis of a multidrug-resistant environmental isolate (DL4.3) and clinical isolate (EspIMS6) of *Enterobacter* spps. stated that the clinical isolate possessed an increase in capsular and extracellular polysaccharides. Also, the clinical isolate conferred resistance towards antibiotics, toxic compounds and bacteriophages, thus representing the role of the bacteriophages in enhancing the pathogenicity of the bacterial strain [106]. Furthermore, *E.r cloacae* (M12X01451), isolated from patients with mild diarrhea, produced subtype Stx1, encoded by a bacteriophage. The stx1-a and stx1e encoding bacteriophage induced from *E. coli* O157:H7 strain RM8530 and M12X01451, respectively, can infect other commensal *E. coli* and *E. cloacae* strains transitioning them to a pathogenic state. The probability of double lysogeny in the commensal bacteria was also demonstrated, indicating horizontal transfer of the prophages, which may induce pathogenicity in the commensal bacteria [107].

In brief, bacteriophages play important role in the pathogenicity of ESKAPE pathogens. Hence, identification and cost effective detection of molecular signals inducing the lysogenic cycle of the bacteriophages need further investigations.

5. Conclusion

Due to the rise in AMR, treating ESKAPE infections with antibiotics is a severe concern. Thus, investigations in the potential use of bacteriophages for therapeutic purposes have gained major interests. The current evidence on the use of mono-bacteriophage therapy, cocktails of bacteriophages, and bacteriophage isolated proteins state that bacteriophage therapy is an effective and inexpensive strategy to combat ESKAPE infections. When working with high doses of bacteriophages, the delivery mechanism to the infection site is needed to be investigated for efficient bacteriophage therapy. Also, the exact ratio of bacteriophages during cocktail treatment is to be determined earnestly. Several clinical trial studies have been performed for bacteriophage therapy, although further molecular data and regulatory policies are required to offer a robust case for clinical use. Moreover, personalized bacteriophage therapy should be given utmost priority, since the human body reacts differently towards different molecules.

Simultaneously, several findings suggest the role of bacteriophage in initiating or enhancing the pathogenicity of ESKAPE pathogens by several mechanisms like horizontal gene transfer. Studies have provided evidence that bacteriophage can disseminate virulence factors coding for adhesion, invasion, and immune evasion due to the horizontal gene transfer mechanism. The inter-relation of bacterial virulence genes with bacteriophages might indicate evolutionary changes, a research gap that needs to be explored. Such a dual role of bacteriophage may grave a threat globally. Therefore, it is crucial to assure safety by unraveling the basic mechanism of bacteriophage therapy and bacteriophage pathogenicity. A better understanding of the interactions between bacteriophage, bacteria and humans, before implementing bacteriophage therapy, is an urgent need.

Moreover, the important aspect of being studied is targeting and recognizing ESKAPE pathogens as host by bacteriophages and influencing various host cell receptors leading to the activation of several signaling pathways. The specific binding between the bacteriophage and the host bacterial cell is crucial for developing novel effective therapeutics.

Therefore, the preliminary evidence of enhanced efficacy of combination therapy indicates that further exploration of bacteriophage cocktails, bacteriophage proteins, bio-engineered bacteriophage, with and without antibiotics in the *in vivo* models are required to address the global challenges of AMR.

6. Expert opinion

To utilize the potential of bacteriophages in anti-infective therapy, it is imperative to understand the role of bacteriophages in the transfer of virulence attributes. Several bacteriophages have been reported to inhibit bacterial infections and are used in some countries as therapeutics, but it has not yet become a standard medical strategy.

The biological safety of bacteriophages is one of the major challenges that need attention. Overall, bacteriophage therapy has tremendous potential; however, to avoid associated risk, selection of the perfect bacteriophage is essential. Bacteriophage therapy is old but again gaining interest due to the rapid increase in the rate of antibiotic resistant infection and lack of effective antimicrobials. Due to the increasing incidences of antibiotic resistant infection, the global bacteriophage therapy market is expected to grow rapidly from 2020 to 2027.

The lysogenic bacteriophage DNA multiply along with bacterial DNA. Integrated bacteriophage DNA can be excised during the lytic phase. Moreover, in this process bacteriophage may acquire some bacterial genes and transfer them to bacteria to which they infect. The use of obligate lytic bacteriophage is essential to avoid the dissemination of antibiotic resistance from bacteriophage to bacteria. The obligate lytic bacteriophage can be synthesized by genetic modification using gene editing systems like CRISPR cas-9. There are multiple reports on the bacteriophage encoding antibiotic resistance genes (ARGs) but the precise contribution of the bacteriophages to transfer ARGs is not yet fully understood. Also, environmental factors inducing prophage state need to be identified.

In addition to the rapid methods for the screening of non-toxic bacteriophage and bacteriophage without antibiotic resistance genes, we need to use advanced biotechnological tools to engineer new synthetic bacteriophage with minimal non-essential components. In order to assure selectivity of the bacteriophages, synthetic bacteriophage or bacteriophage cassette can be combined with the receptor binding proteins (RBPs). Most of the bacterial infections are polymicrobial in nature. Hence, to target a wide range of pathogenic strains at the infection site, multivalent synthetic bacteriophages can be developed by combining multiple RBPs. The use of micro viruses is more advantageous as they are small in size and non-transducing. However, extensive investigations will be required to develop synthetic bacteriophages as we have limited information in this field.

Regarding the specificity of bacteriophage therapy, it is important to achieve specificity between near narrow and near broad. Narrow specificity with mono-bacteriophage therapy may have the disadvantage of not being effective against other strains of the same species. However, broad range bacteriophage therapy consisting of bacteriophage cocktails may affect the normal beneficial microbiome. Hence, bacteriophages without cross activity against other species but effective against all strains of target bacterial species should be preferred. The research attempts for the development of screening tools for the rapid selection of bacteriophages with optimal selectivity are highly required for the successful eradication of the infection without side-effects.

The personalized bacteriophage therapy employs the creation of a bank of bacteriophage as "Active Pharmaceutical ingredients" and depending on the patient's infection effective bacteriophages are screened from the bank and formulated to develop medicine. The personalized medicine model has been successful in the US and Europe. However, total treatment cost and efficacy in the “Personalized medicine model” and “Fixed drug models” of bacteriophage therapy have not been comparatively studied.

A recent study demonstrated that bacteriophage cocktail therapy involving the simultaneous administration of bacteriophages is effective compared to sequential bacteriophage administration (Wright et al., 2019). However, the underlying mechanism is not much explored in relation to the ESKAPE pathogens. Another study displayed that the lytic capacity of the bacteriophage partially depends on the inter-relationship between bacteriophage and bacteria (Principi et al., 2019). Therefore, it is essential to understand the fundamental processes involved in the bacteriophage-bacteria interaction and its connection with the antimicrobial resistance

Bacteriophage prior adapted with bacterial strain has shown increased efficacy. However, further studies in a similar line need to be carried out with ESKAPE pathogens for the development of effective treatment strategies against these pathogens. To avoid the emergence of bacteriophage resistant bacteria multispecific bacteriophage cocktail is recommended; however, the effect of a multispecific bacteriophage cocktail on the normal microbiome is not studied in detail.

It has been argued that when bacteria become bacteriophage resistant, they become sensitive to antibiotics. Therefore, bacteriophage therapy in combination with antibiotics is more effective than bacteriophage therapy alone.

Earlier most of the ESKAPE pathogens were considered as extracellular. However, recent findings revealed the intracellular existence of the ESKAPE pathogens. Hence, the efficacy of bacteriophage therapy against intracellular ESKAPE pathogens needs further investigation.

Moreover, the mechanism of bacteriophage infection, replication and how bacteria protect themselves from bacteriophage treatment is yet elusive. To assure the safety of the therapeutic bacteriophage it is important to understand the function of each gene and protein of the bacteriophage. However, many bacteriophages are not yet sequenced and the molecular function of the protein is not fully understood. Hence, collaborative efforts by biochemists, microbiologists, molecular biologists and clinicians are required to win the battle against pathogens. Governments and public health organizations need to finance basic research on bacteriophage therapy, to develop comprehensive database of the viruses and their target bacteria.

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Table 1. Bacteriophages in the treatment of ESKAPE infections

Pathogen	Strain	Disease	Type of therapy	Details of therapy	Study model	Route of injection for in vivo models	Treatment time after challenge	Efficacy	Additional findings	Reference	R
Enterococcus	*E. faecalis* V583 (ATCC 700802) Planktonic and biofilm		Bacteriophage/ bacteriophage cocktail therapy	Bacteriophage EFDG1	*In-vitro* human root canal model		7 days	5-log reduction of bacteria within biofilm	EFDG1 genome does not possess toxic genes	[23]	[2
	E. faecalis and *E. faecium* biofilm	Wound site infection		Bacteriophage vB_EfaS-Zip or vB_EfaP-Max (Max)	*In vitro*		3h	1.5-2 log CFU/ml reduction in bacterial load within biofilm	Genome of both bacteriophage does not contain genes coding for lysogeny	[24]	[2
	VRE V583	Human dentin root segments		Engineered bacteriophage φEf11/φFL1C (Δ36)	In vitro		3 days of bacteriophage therapy	99%		[27]	[2
	VREF			Bacteriophage EFp29	Mice model		1h	100%		[28]	[2
	Clinical isolates of *E. faecalis*			Bacteriophage HEF13	In vitro infection models of human			100%	Bacteriophage HEF13 possess wide range of temperature (4-60 °C) and pH (pH 3-12) stability	[29]	[2

					dentin						
	E. faecalis Biofilm			Bacteriophage EF1TV	In vitro			9-68% reduction in the biofilm development		[25]	[2
	E. faecalis (V583) biofilm			EFDG1 and EFLK1 in 1:1 ratio	In vitro fibrin-clot infection model		6h	6 log reduction in the CFU/ml	Bacteriophage cocktail exhibited higher efficacy compared to individual bacteriophage and maximum activity was contributed by EFLK1	[26]	[2
	E. faecalis EF24 or VRE2	Endophthalmitis		ΦEF24C -P2	Mouse model	Intra-vitreous	6-12h	Significant reduction in the bacterial load and attenuation of clinical signs	Intravitreous administration of ΦEF24C -P2 neither induced inflammatory response nor caused retinal toxicity	[34]	[3
	E. faecalis	Clinical isolate-from orthopedic implant infections		Bacteriophages vB_EfaS_LM99	In vitro		2h	99%		[30]	[3
			Combination therapy	Bacteriophage cocktail (EFDG1 and EFLK1) with	Female mice ICR (CD-1) model	Intra-peritoneal	0h-1h	100%		[31]	[3

				antibiotic (12.5 mg/kg of ampicillin)							
	E. faecalis biofilm			Bacteriophage EFLK1 (1.2×10^8 PFU/well) and vancomycin (0.015 mg/mL)	In-vitro		72h biofilm	7 log reduction in the planktonic state and reduction to undetectable level in the biofilm state	Combination therapy showed higher efficacy as compared to individual bacteriophage or antibiotic treatment	[108]	[1
	E faecalis	Root Canal infection		Bacteriophage EFDG1 and EFLK1 formulated with Kolliphor P 407 (poloxamer)	Rat model		72h	99% reduction in the bacterial load (CFU/ml)	Combination therapy showed greater reduction in the bacterial load compared to bacteriophage treatment alone	[33]	[3
	E. faecium 4P-SA	Clinical sample	Bacteriophag protein therapy	LysEFm5 lysin protein purified from IME-EFm5	In vitro		60min	100 % reduction in the growth of all clinical strains at 16 to 128 µg/ml	Protein showed broader activity compared to bacteriophage	[109]	[1
	Enterococcus			VD13 bacteriopha				5.50 ± 1.76 $\Delta OD_{600\ nm}$	Bacteriophage protein showed maximum activity in the pH	[110]	[1

	faecalis			ge endolysin				min−1 mg−1 protein	range 4-8 (peak activity at pH 5), in the presence of calcium and in the absence of NaCl		
	Enterococ ci	E. faecium 35667 and E. faecalis MMA1		Chimeric lysin Ply187-V12C	In-vitro			100% reduction in the growth of bacteria at 5 µM	Ply187-V12C was engineered by combining catalytic domain -Ply187N from the bacteriophage lysine Ply187 and the cell binding domain V12C -146aa-314aa of the bacteriophage lysin PlyV12) and is effective against different pathogenic genera	[111]	[1
	E. faecalis	Endodonti cs		The chimeric lysine-ClyR				>90%		[32]	[3
S. aureus	MRSA isolates/ ATCC strains		Bacteri ophage / bacterio phage cocktail therapy	Mono-bacteriopha ge therapy (isolates)/ Bacteriopha ge cocktail (M1, M1M4, CJ17, CJ12)	In vitro (abiotic surface)		4h	90-100%	Bacteriophage isolates depicted lytic activity in both liquid culture and in decontaminating abiotic surfaces involved in nosocomial transmission.	[36]	[3
	IPLA1/ IPLA16	Food poisoning		Mono-bacteriopha ge therapy (phiIPLA-ROD)	In vitro (biofilm assay)		4h	2.4-4.2 log reduction of bacteria growth	Bacteriophage treatment on *S. aureus* mixed-species biofilms varies depending on the accompanying species and the infection conditions.	[72]	[7
	ATCC 43300	Skin and soft tissue		Transfersom e-entrapped	Rat	Intramuscul ar	7-10 days	2-5 log reduction in	Transfersomes as delivery agents enhance the stability	[112]	[1

		infections		phage cocktail (MR-5 and MR-10)				the bacterial growth	and in vivo persistence of encapsulated phages.		
	Clinical isolate			Bacteriophage K	In vitro		3-4 days	83%	Bacteriophage K impacted MHC-I/II and CD80/CD86 protein expression suggesting that phage K does not independently affect cellular immunity.	[113]	[1
	Clinical isolate	Hospital acquired infections		Bacteriophage based drug Staphal	In vitro		24-48h	80-100%	Bacteriophages destroy planktonic cells effectively. For biofilm destruction, bacteriophage concentration needs to be at least ten times higher than that provided.	[41]	[4
	Clinical isolate	Ortheopedic implant-associated infection		Bacteriophage LM12	In vitro		2-24h	91-99%	Bacteriophage therapy studies depict that under certain conditions, bacteriophages are effective against bacterial population to a level that can allow activation of the host immune response to successfully clear the infection.	[30]	[3
	Clinical isolate	Diabetic foot ulcer with osteomylitis		Bacteriophage Sb-1	Case study	Soft tissue injection	7 weeks	Successful qualitative treatment		[114]	[1
	Clinical isolate	chronic wound		Bacteriophage cocktail	Clinical study	Topical application	21 days	40-60%	Significant improvement was observed with no signs of infection after 3 to 5 doses of	[115]	[1

									topical bacteriophage therapy		
	Clinical isolate	Recalcitrant chronic rhinosinusitis		Bacteriophage Cocktail AB-SAO1 (Sa83+ Sa87+J-Sa36)	Phase 1 Clinical trial	Intranasal	14 days	2/9 patients had eradication of infection.	Intranasal bacteriophage treatment was efficient, with no serious adverse events or deaths.	[37]	[3
	Clinical isolate			Bacteriophage Cocktail AB-SAO1 (Sa83+ Sa87+J-Sa36)	In vitro and mouse model	Intranasal	24h	94-95%	The preclinical data supportes the production of AB-SA01 under good manufacturing practices and phase 1 clinical studies.	[38]	[3
	Clinical isolate	Endocarditis		Bacteriophage cocktail AB-SAO1 (Sa83+ Sa87+J-Sa36)	Case study	Intravenous	14 days	Successful qualitative treatment		[40]	[4
		Orthopedic implant related infection		Sta Bacteriophage cocktail	In vitro (biofilm assay)		8h	98-100%	Compared with cefazolin treatment, reduction in *S. aureus* biofilms growing on titanium surfaces after exposure to a bacteriophage cocktail was significant.	[42]	[4
	Clinical isolate	Bacteremia		Bacteriophage Cocktail AB-SAO1 (Sa83+ Sa87+J-Sa36)	single-arm non-comparative trial	Intravenous	14 days	Successful qualitative treatment	No adverse reactions were reported, and the data supports the use of AB-SA01 against severe *S. aureus* infections, including infective endocarditis and	[39]	[3

									septic shock.		
	Isolate		Combination therapy	Mono-bacteriophage therapy (isolates)/ Combination therapy (clindamycin 8mg/kg)	Mice model	Intravenous	24-72h	75-100%	Mono-bacteriophage therapy was more effective (100%) than with clindamycin (62-87%) or combination treatment (75-90%).	[43]	[4
	ATCC 43300	Hospital acquired infections		Monophage therapy (Sb-1)/ Combination (Sb-1+fosfomycin/ rifampin/ vancomycin / daptomycin/ ciproflaxin)	In vitro		3-24h	75-90%	Sb-1 bacteriophage lyses *S. aureus* planktonic cells, but it does not eradicate the biofilm.	[44]	[4
	Newman	Skin infections		Bacteriophage PYO+ antibiotics (gentamycin / oxacillin/ vancomycin / tetracyclin/ ciproflaxin/ daptomycin/ erythromycin/linzoid)	In vitro		48h	30-50 fold reduction in bacteria	Gentamycin and Oxacillin were more effective during sequential with bacteriophage than when added at the same time.	[116]	[1

	HG001	Osteoblast infection		Bacteriophage cocktail (PP1493, PP1815, PP1957)+ antibiotic (1.5mg/L vancomycin / 0.016mg/L rifampin)	In vitro		24h	3.1-3.6 log reduction in bacteria	Bacteriophage penetration into osteoblasts suggested a *S. aureus*-dependent Trojan horse mechanism for internalization. Therapy was ineffective against bacteria internalized in osteoblasts.	[117]	[1
	USA300	Pneumonia	Bacteriophage protein therapy	Lysin LysGH15 from bacteriophage GH15 + apigenin	In vitro & mouse model	Lys GH15-intranasal, apigenin-subcutaneous	1h	100%	Apigenin (natural flavonoid) inhibits transcription of *hla* and *agrA* encoding hemolysin in *S. aureus* playing anti-inflammatory role where disintegration of *S. aureus* by LysGH15 could release toxins leading in inflammation.	[45]	[4
	USA300	Infected skin wound		Lysin LysGH15-api-Aquaphor (ointment)	In vitro and mouse model	Topical application	18-96h	90-100%	Ointment also reduced the expression of TNF-α, IL-1 and IFN-γ and accelerated wound healing in the mouse model.	[46]	[4
	Clinical isolate	Atopic dermatitis		Staphefekt SA.100 (recombinant bacteriophage endolysin containing cream)	Case study	Topical application	3-4 weeks	Successful qualitative treatment	Staphefekt targets both MSSA and MRSA but does not affect commensal skin microbes due to its specific action of mechanism.	[118]	[1

	USA300	Bacteremia		Lysin P128	Rat	Intravenous	14 days	54-100%	Treatment with P128 by intravenous bolus administration or infusion via the jugular vein resulted dose-dependent survival of rats.	[47]	[4
	N315			Lysin PlyV12 (V12CBD)	In vitro and in vivo (mouse)	Intraperitoneal	24h	2.4-23.4-fold reduction	V12CBD suppresses invasion and expression of virulence factors in *S. aureus* and activates macrophages and enhance phagocytosis against *S. aureus*.	[119]	[1
	Isolate			Lysin Lys-phiSA012	In vitro		30-45 min	80-90%	Presence of Ca2+ and MZn2+ enhances the lytic activity.	[49]	[4
	ATCC and CECT strains			Bacteriophage and endolysin vB_SauM-LM12 (LM12)	In vitro		2-24h	90-100%	LM12 endolysin was active against stationary-phase cells and suspended biofilm cells than against intact and scraped biofilms, suggesting that cellular aggregates protected by the biofilm matrix reduced protein activity.	[50]	[5
	ATCC strains	Bacteremia		Lysin SAL200+ antibiotics (25mg/kg vancomycin / 200mg/kg nafcillin)	In vitro and in vivo (mouse and *Galleria mellonella*)	Mouse-intravenous and intraperitoneal, *G. mellonella*-injected into the larva	30min-72h	3 log reduction	SAL200 acts synergistically with antibiotics in vitro.	[120]	[1

						through the last left pro-leg					
	LAC USA300	Pneumonia		Lysin Sal200	Mouse	Intranasal	1h	90-95%	SAL200 lead to low mortality without a significant increase in inflammatory cytokines.	[48]	[4
K. pneumoniae	MTCC 5832	Pneumonia	Bacteriophage / bacteriophage cocktail therapy	Liposome entrapped bacteriophage KPO1K2	Mouse	Intraperitoneal	24h	2-5 log reduction in bacteria	Liposome mediated bacteriophage delivery caused a significant reduction in proinflammatory cytokine levels and an increase in the levels of anti-inflammatory cytokines.	[121]	[1
	B5055	Burn wound infection		Bacteriophage cocktail (Kpn1, Kpn2, Kpn3, Kpn4 and Kpn5)	In vitro and Mouse model	Topical application	1 day	3 log reduction in bacteria	Strategy is effective for treating wound infections in individuals showing resistance to antibiotic therapy.	[122,123]	[1
	KP/01	Wound infection		Bacteriophage ZCKP1	In vitro		3h	2 log reduction in bacteria	ZCKP1 can reduce *K. pneumoniae* in both planktonic and biofilms and is extremely stable over abroad range of pH and temperatures.	[124]	[1
	12C47			Bacteriophage vB_Kpn_F48	In vitro		2-4h	0.5-1 log reduction in bacteria	Physiological characterization suggested that vB_Kpn_F48 depicted narrow host range, and high stability to temperature and pH variations.	[125]	[1

	kp235			Bacteriophage cocktail (ECP311, KPP235, ELP140)	In vitro and in vivo *(G. mellonella)*	Injected into the larva through the last left pro-leg	48h	100%	In vivo models required multiple bacteriophage doses for effective treatment.	[52,53]	[5
	ShA2	Hospital acquired infections		Bacteriophage TSK1	In vitro		24h	99-100%	TSK1 produces depolymerase enzyme which plays a role in *K. pneumoniae* capsular polysaccharide and biofilm disruption.	[54]	[5
	Clinical isolate	Urinary tract infections		Monophage therapy/ Cocktail (117 and 31)	In vitro		8h	80%	In LB cultures, cocktail showed significantly higher antimicrobial activity than bacteriophage alone, although this was not observed in urine samples.	[126]	[1
	Clinical isolate	Non-healing ulcers		Monophage therapy/ Cocktail (Isolates)	Case study	Topical application	90 days	812%	Post bacteriophage therapy, hemoglobin and lymphocytes level increased significantly.	[127]	[1
	Clinical isolate	Neonatal sepsis		Bacteriophage cocktail	Mouse	Intraperitoneal	20 days	100%	Polyvalent bacteriophage preparations was successful in treatment of polymicrobial animal model for bacteremia.	[55]	[5
	MTCC 109	Pneumonia		Bacteriophage VTCCBPA43	Mouse	Intranasal	96h	80-100%	VTCCBPA43 was stable at high temperature and most active at pH 5. Also exhibited a narrow host range.	[51]	[5
A. baumannii	G7 and T-10	Wound infection	Bacteriophage	Bacteriophage vB-	Rat	Topical application	4-5 days	100%	vBGEC_Ab-M-G7 depicted short latent period and wide	[56]	[5

			/ bacteriophage cocktail therapy	GEC_Ab-M-G7					host range. Also, resistance to chloroform and thermal and pH stability was observed.		
	Clinical isolate			Bacteriophage Bϕ -C62	In vitro, in vivo	Intranasal	7 days	100%	Survival rates differed according to bacteriophage MOI.	[57]	[5
	Clinical isolate	Pneumonia		Bacteriophage vB_AbaM-IME-AB2	In vitro		12h	88.50%	Disinfection assay depicted that the bacteriophage lysed *A. baumannii* cells suspended in blood/ metal surfaces.	[128]	[1
	Clinical isolate	Pneumonia		Bacteriophage vB_AbaM-IME-AB2	Mouse	Intranasal	144h	100%	Microcomputed tomography depicted inflammation reduction in lung of mice on bacteriophage treatment.	[58]	[5
	Clinical isolate	Necrotizing pancreatitis		Bacteriophage cocktail ϕPC/ ϕIV/ ϕIVB	Case study	Intravenous and intracavitary	7-8 weeks	Successful qualitative treatment	Development of personalized bacteriophage-based therapy for a 68YO diabetic patient with complicated MDR *A. baumannii* infection.	[129]	[1
	AB1	Burn wound infection		Bacteriophage Abp1	In vitro and in vivo (mouse)	Subcutaneous or direct pipetting into the wound	6 days	100%	Abp1 can lyse all *A. baumannii* at low MOI, suggesting that a small number of bacteriophage may be sufficient for effective bacteriophage therapy.	[60]	[6
	Clinical isolate	Lung infection		Bacteriophage cocktail (SH-Ab 15519, SH-Ab 15497, SH-Ab 15708, and SH-Ab	Mouse	Intranasal	14 days	87.50%	In silico analysis depicted that bacteriophage SH-Ab15519 is a novel bacteriophage with no virulence or antibiotic resistance genes.	[130]	[1

				15599)							
	KM18	Bacteremia		Bacteriophage ϕkm18p	Mouse	Intraperitoneal	14 days	100%	Bacteriophage therapy decreased the levels of TNF-α and IL-6.	[59]	[5
	Clinical isolate	Nasal infection		Bacteriophage cocktail (PBAB08 and PBAB25)	Mouse	Intranasal	7 days	2.3fold higher survival rate	20% increase in IgE production was observed in intraperitoneal injection of the bacteriophage cocktail	[131]	[1
	Clinical isolate			Bacteriophage WCHABP1 and WCHABP12	In vivo (*G. mellonella*)	Proleg injection	96h	75%	In silico study for WCHABP1 and WCHABP12 showed significant DNA sequence similarity with bacteriophages of the Ap22virus.	[132]	[1
	Clinical isolate	Pneumonia		Bacteriophage Bϕ-R2096	In vivo (*G. mellonella* and mouse)	*G. mellonella*- proleg injection, Mouse- intranasal	*G. mellonella*- 24h, mouse- 12 days	*G. mellonella*- 50%, mouse- 100%	Bacteriophage inhibited bacterial growth in dose-dependent manner.	[61]	[6
	OXA-23, AmpC and clinical isolates	Bacteremia		Bacteriophage cocktail (vB_AbaM_3054 and vB_AbaM_3090)	In vitro and In vivo (*G. mellonella* and mouse)	*G. mellonella*- proleg injection, mouse- intraperitoneal	7 days	100%		[62]	[6

	Clinical isolate	Bacteremia		Bacteriophage cocktail (vB_AbaS_D0 and B_AbaP_D2)	Invitro, In vivo (mouse)	Intraperitoneal	36h	90-100%	Bacteriophage cocktail was more efficient than mono-bacteriophage therapy	[63]	[6
P. aeruginosa	Clinical isolate	UTI	Bacteriophage / bacteriophage cocktail therapy	KT28 and KTN6	In vitro		4h		Inactivated KT28 is equally efficient as that of active KT28. Reduction in pyocyanin synthesis is also reported.	[133]	[1
	Clinical isolate			Bacteriophage cocktail (DL52, DL52, DL60, DL62, DL64, DL68)	In vitro		22– 48h	95% in static model	In the dynamic model biofilm is dispersed after 48h of exposure to bacteriophage.	[64]	[6
	Clinical isolate (PA45291, BC09007)	Bacteraemia, cystic fibrosis		Cocktail bacteriophage	*G. mellonella*	Injected into haemolymph	15h	85-100%	In first model bacteriophage is introduced 2h prior infection and in second 2h post infection. In both cases 30h post infection mortality was observed.	[65]	[6
	(ATCC 15692), nonCF00and CF708	Cystic Fibrosis		KTN4	Airway Surface Liquid infection model and wax moth larvae model	Injected into haemolymph	72h	4-7 log reduction in bacteria	The ASL model gives the basis of further in vivo evaluation. The bacteriophage exhibits anti-biofilm potential.	[134]	[1

	Clinical isolate	Chronic rhinosinusitis		Cocktail bacteriophage (Pa193, Pa204, Pa222, Pa223)	Animal model – Sheep	Frontal trephine flushes	7 days	2 log reduction in bacteria		[135]	[1
	Clinical isolate	Endocarditis		Cocktail PP1131	Rat	Continuous or bolus injection	6h	2.3-3 log reduction in bacteria	The combination of bacteriophages with ciprofloxacin was synergistic. Bacteriophage mediated killing was correlated with cytokines	[136]	[1
	LESB65	Short term acute respiratory infection		Bacteriophage PELP20	Murine model	Intranasal	6 days	3 log reduction in biofilm		[137]	[1
	Clinical isolate	Acute kidney injury		Cocktail BFC1	Case study	Intravenous	10days	Successful qualitative treatment	Patient recovered during treatment, but died due to cardiac arrest after 4 months due to sepsis.	[138]	[1
	Clinical isolate	Sepsis and bacteremia		Bacteriophage OMKO1	Case study	Topical application	24 h	Successful qualitative treatment	Bacteriophage alone or in combination with ciprofloxacin or ceftazidime reduced biofilm density.	[139]	[1
	Clinical isolate	Acute respiratory infection		Cocktail (PYO2, DEV, E215, E217, PAK_P1 and PAK_P4)	Mice and *G. mellonella*	Mouse-Intranasal, *G. mellonella* – in haemolymph	48h	Mice-100%, *G. mellonella*-65%	Cocktail more efficient than individual bacteriophage on biofilm in mice.	[67]	[6
	Clinical isolate	Wound infections		Cocktail PP1131	Case study	Applied on wound –	14 days	Successful qualitative	Longer time period is required to lyse bacteria as compared to	[140]	[1

		and sepsis				cutaneous		treatment	standard care i.e., Sulfadiazine silver		
	Clinical isolates	Infection – biofilm on medical devices	Combination therapy	Amikacin with bacteriophage / Meropenem with bacteriophage	In vitro study		24h	Amikacin + bacteriophage (86.6%), meropenem + bacteriophage (73.3%)	Amikacin – bacteriophage combination is recommended over meropenem to eradicate biofilm	[141]	[1
	Isolates			NP1 AND NP3	In vitro		48h		Bacteriophage treatment in combination with antibiotic shows synergy and reduces amount of antibiotic required.	[142]	[1
Enterobacter	*E. cloacae* laboratory isolate	Urinary tract infections	Bacteriophage / bacteriophage cocktail therapy	E2,E3,E4 separately & cocktail	In vitro study		24 hours	2-4 log reduction	Maximum of bacterium inactivation occurs by the E-4 bacteriophage after 6 h	[68]	[6
	(*E. cloaae*) Clinical isolate	Lower respiratory tract infection, Urinary tract infection		MJ2	In vitro study		4h	2.8-3.5 log reduction in bacteria	The bacteriophage shows stability over a range of temperature and pH. In the exponential growth phase planktonic cells are more sensitive to MJ2.	[143]	[1
	E. cloacae el140	Bacteraemia		Bacteriophage- ELP140 & bacteriophage cocktail (ECP311, KPP235 and	*G. mellonella*	Injected into haemolymph	96h	100%	100% efficacy is obtained with 3 doses of bacteriophage at 6h interval.	[52]	[5

				ELP140)							
	(*E. cloaae*) Clinical isolate	Bacteremia		Cocktail bacteriophage KL2 (*Escherichia* virus myPSH2311, *Klebsiella* virus myPSH1235, *Enterobacter* virus myPSH1140)	In vitro study		4h	2-3 log reduction in bacteria	Bacteriophage cocktail shows similar activity as bacteriophages alone against meropenem and colistin resistant bacteria.	[53]	[5